CHEMICAL KINETICS

Introduction : Chemical kinetics is a branch of physical chemistry which deals with the rates of chemical reactions, the mechanism by which they proceed and the influence of different factors like pressure, temperature, concentration, volume etc on the rate of the reaction.

Molecularity : It is defined as the total number of reactant molecules taking part in the chemical reaction, as represented by a balanced chemical equation.

Eg: (1) If one reactant molecule is taking part, then that reaction is called unimolecular reaction.

$$A \longrightarrow \text{Products} \quad \text{or} \quad H_2O \longrightarrow H_2 + \frac{1}{2} O_2$$

(2) If two reactant molecules take part then that reaction is called bimolecular reaction.

$$2A \longrightarrow \text{Products} \quad \text{or} \quad 2HI \longrightarrow H_2 + I_2$$

(3) If three reactant molecules take part then that reaction is called trimolecular reaction.

$$3A \longrightarrow \text{Products} \qquad \text{or} \qquad 2NO + O_2 \longrightarrow 2NO_2$$

Order of the reaction: The order of the reaction depends on ‘ rate equation’ of the reaction.

Definition: *Order of a reaction is defined as the sum of the powers to which the concentration terms of the various reactants are raised to the rate equation.*

Eq: 1) For a reaction of the type $A \longrightarrow$ Products

Rate equation is $\frac{dc}{dt} \alpha [A]^1$ Or $\frac{dc}{dt} = k [A]^1$

Since the concentration term is raised to the power one, the order of the reaction is one.

2) Similarly, $2A \longrightarrow$ Products

Therefore, $dc/dt \ \alpha \ [A]^2$ or $dc/dt = k \ [A]^2$ therefore, Order = 2

3) Similarly, $A + B \longrightarrow$ Products

Therefore , dc/dt =$[A]^1[B]^1$ or dc/dt =$[A]^1[B]^1$, Order=1+1=2

4) Similarly, $3H_2 + N_2 \longrightarrow 2NH_3$

$\frac{dc}{dt}$ α $[H_2]^3[N_2]^1$ OR dc/dt =k $[H_2]^3[N_2]^1$

Therefore order=3+1=4

Note: i. Molecularity of a reaction is always a whole number and cannot be zero or fraction.**ii.** Order of reaction can be zero, fraction or whole number.

Pseudo uniorder reactions: A chemical reaction whose order is one but the molecularity is two or more is called pseudo uni-order reaction.

Eg: $CH_3COOCH_3 + H_2O \longrightarrow CH_3COOH + CH_3OH$

Molecularity = 2 , order = 1

Rate equation, dc/dt α $[CH_3COOCH_3]^1$ $[H_2O]^1$

= k $[CH_3COOCH_3]^1$ $[H_2O]^1$

In this case water is present in large quantity and its concentration remains almost constant. Therefore to find the order only the concentration of CH_3COOCH_3 is taken into consideration .This type of reactions are called pseudo uniorder reactions.

Differential and Integrated rate equations:

Order	Example	Differential rate Equation	Integral rate equation
1	A → Product	dc/dt=k(a-c)	$k=\frac{2.303}{t}\log\frac{a}{a-c}$
2	2A → Product	$dc/dt=k(a-c)^2$	$k=\frac{1}{t}\frac{c}{a(a-c)}$
2	A+B → Product	dc/dt= k(a-c)(b-c)	k=2.303 log (a-c)b
n	nA → Product	$dc/dt= k(a-c)^n$	$k=\frac{1}{t(n-1)}\frac{1}{a^{n-1}}[2^{n-1}-1]$

Second order reactions:

Definition : *A reaction in which the reaction rate is determined by the variation of the concentration of two reactants.*

Eg: Second order reaction can be of three types

(i) Two identical reactants molecules taking part in the reaction

$2A \longrightarrow$ Product

$2HI \longrightarrow H_2 + I_2$

(ii) Two different reactant molecules of different initial concentrations taking part in the reaction.

$A + B \longrightarrow$ Products

$H_2 + I_2 \longrightarrow 2HI$

(iii) Two different reactant molecules of identical initial concentration taking part in the reaction.

$A + B \longrightarrow$ Products where [A] =[B]

$H_2 + I_2 \longrightarrow$ 2HI where $[H_2]=[I_2]$

Rate equation for a second order reaction in which two identical reactants (2A) or two different reactants A and B with equal initial concentration ([A]=[B] OR a=b):

Let us consider a second order reaction of the type,

$2A \longrightarrow$ Product OR

$A + B \longrightarrow$ Product where [A]=[B]

Let initial concentration of reactants= 'a' mol/dm^3

Let the amount of reactants reacted at the time 't' = 'c' mol/dm^3

The amount of reactants left at the time, 't'= (a-c) mol/dm^3

Rate of the reaction, $\frac{dc}{dt} \alpha [A]^2$

$$\frac{dc}{dt} = k [A]^2$$

, $$\frac{dc}{dt} = k (a-c)^2 \quad \text{.....(1)}$$

Where 'k'is called proportionality constant. Here 'k'is also called as Rate Constant or velocity constant.

Rearranging the equation (1), we have $\frac{dc}{(a-c)^2} = k\,dt$ ..... (2)

To find the value of 'k', the equation (2) is integrated on both sides,

$\int \frac{dc}{(a-c)^2} = k\int dt$

$(a-c)^{-1} = kt + X$ ..... (3)

Where 'X' is integration constant and to find value of 'X' ,let us consider that when t=0, c=0 . Substituting the value of t=0 and c=0 in equation (3) we get

$$\frac{1}{a-0} = k\times 0 + X$$

$$\frac{1}{a} = X \quad \text{..... (4)}$$

Substituting $X = \frac{1}{a}$ in equation (3), we get

$$\frac{1}{a-c} = kt + \frac{1}{a}$$

$$kt = \frac{1}{a-c} - \frac{1}{a} \Rightarrow k = \frac{a-(a-c)}{a(a-c)} = \frac{c}{a(a-c)}$$

$$k = \frac{1}{t}\times\frac{c}{a(a-c)} \quad \text{..... (5)}$$

Equation (5) gives the *integral form of velocity or rate constant* of a second order reaction in which [A]=[B].

<u>Unit of second order rate constant</u>

$$k = \frac{1}{t}\times\frac{c}{a(a-c)}$$

$$K = \frac{1}{time} \times \frac{mol\ dm^{-3}}{moldm^{-3}\ mol\ dm^{-3}}$$

$$k = \frac{1}{time}\times\frac{1}{\text{moldm}-3}$$

$= \boldsymbol{time^{-1}mol^{-1}dm^3}$ OR $\mathbf{sec^{-1}mol^{-1}litre}$

Problems: (1) A second order reaction with two reactants is started with 0.1 M concentration each. It is 10% completed in 30 minutes. Calculate the rate constant. [ans: 3.7×10^{-2} $min^{-1}mol^{-1}dm^3$]

(2) A second order reaction in which a=b is 30% completed in 600s . Calculate the time required for the completion of 60% of the reaction.
[ans:7.14×10^{-6} $min^{-1}mol^{-1}dm^3$ and 2100s]

Rate expression for a second order reaction of the type A+B → Produdcts in which [A]≠[B] OR (a≠b):

Let us consider a second order reaction of the type,

A+ B ⟶ Product where [A]≠[B]

Let initial concentration of reactant A = 'a' mol/dm^3

Let initial concentration of reactant B = 'b' mol/dm^3

Let the amount of A reacted at the time 't' = 'c' mol/dm^3

Let the amount of B reacted at the time 't' = 'c' mol/dm^3

The amount of reactant A left at the time 't'= (a-c) mol/dm^3

The amount of reactant B left at the time 't'= (b-c) mol/dm^3

Rate of the reaction, $\frac{dc}{dt}$ α [A][B]

$$\frac{dc}{dt} \alpha (a-c)(b-c)$$

$$\frac{dc}{dt} = k (a-c)(b-c) \quad \text{.....(1)}$$

Where 'k' is called proportionality constant. Here 'k' is also called as Rate Constant / velocity constant.

Rearranging the equation (1), we have $\frac{dc}{(a-c)(b-c)} = k\,dt$ .....(2)

Resolving the left hand side of the equation (2) into partial fractions, we get

$$\frac{1}{a-b}\left[\frac{dc}{b-c} - \frac{dc}{a-c}\right] = k\,dt \quad \text{..... (3)}$$

To find the value of 'k', the equation (3) is integrated on both sides,

$$\frac{1}{a-b}\left[\int \frac{dc}{b-c} - \int \frac{dc}{a-c}\right] = \int k dt$$

$$\frac{1}{a-b}\left[- \ln (b-c) - [-\ln(a-c)] \right] = kt + X \quad \text{..... (4)}$$

Where 'X' is integration constant and to find value of ' X ', let us consider that when t=0, c=0 . Substituting the value of t=0 and c=0 in equation (4) we get

$$\frac{1}{a-b}\ \{-\ln b - [-\ln a]\} = X \text{ .....(5)}$$

$$\frac{1}{a-b}\ [\ln \frac{a}{b}] = X \text{ .....(6)}$$

Substituting the value of X in equation (5) we get,

$$\frac{1}{a-b}\ \{-\ln (b-c) - [-\ln (a-c)]\} = kt + \frac{1}{a-b}\ [\ln \frac{a}{b}]$$

$$\frac{1}{a-b}\ \{-\ln (b-c) - [-\ln (a-c)]\} - \frac{1}{a-b}\ [\ln \frac{a}{b}] = kt$$

$$\frac{1}{a-b}\ [\ln \frac{a-c}{b-c} - \ln \frac{a}{b}] = kt$$

$$\frac{1}{a-b}\left[\ln \frac{a-c}{b-c} \times \frac{b}{a}\right] = kt$$

$$k = \frac{2.303}{t(a-b)} \log \frac{(a-c)}{(b-c)} \times \frac{b}{a}$$

This Equation gives the integral form of velocity constant of a second order reaction in which **[A]≠[B]** or **a≠b.**

Unit of second order rate constant:

The unit for a second order rate equation is $\boldsymbol{time^{-1}mol^{-1}dm^3}$ OR $\mathbf{sec^{-1}mol^{-1}litre}$. The unit is same for both the type of reactions in which [A]=[B] and [A]≠[B].

Velocity constant or rate constant for n^{th} order reaction

Consider a nth order reaction of the type,

n A ⟶ Products

if 'a'is the initial concentration of the reactant Á' and 'c' is the amount of 'A' undergone reaction at the time 't' then the rate of the reaction can be represented as

$$\frac{dc}{dt} \alpha [A]^n$$

$$\frac{dc}{dt} = k [A]^n$$

,

$$\frac{dc}{dt} = k (a-c)^n$$

$$\frac{dc}{(a-c)^n} = k\, dt \text{ .....(1)}$$

Where ‘k’is called proportionality constant. Here ‘k’is also called as Rate Constant / velocity constant, (a-c) is the amount of A left at time t . the equation (1) is the **differential form of the rate equation for nth order reaction.**

$$k = \frac{1}{t(n-1)} \left\{\frac{1}{(a-c)^{n-1}} - \frac{1}{a^{n-1}}\right\}$$

This equation gives the rate constant for nth order reaction.

Half life period($t_{1/2}$) or ($t_{0.5}$):

Definition: *The time in which the initial concentration is reduced to half of its initial value OR The time in which half of the reaction is completed.*

Relation between Half life period and initial concentration:

1. **For nth order reaction:**

Consider a nth order reaction of the type

$n\,A \longrightarrow$ Products

velocity constant , $k = \frac{1}{t(n-1)} \left[\frac{1}{(a-c)^{n-1}} - \frac{1}{a^{n-1}}\right]$

at half life time, we can substitute $t = t_{1/2}$ and $c = \frac{a}{2}$

$$k = \frac{1}{t\ (n-1)} \left[\frac{1}{(a-\frac{a}{2})^{n-1}} - \frac{1}{a^{n-1}}\right]$$

$$k = \frac{1}{t\ (n-1)} \left[\frac{1}{(\frac{a}{2})^{n-1}} - \frac{1}{a^{n-1}}\right]$$

$$k = \frac{1}{t\ (n-1)} \left[\frac{(2)}{(a)^{n-1}} - \frac{1}{a^{n-1}}\right]$$

$$k = \frac{1}{t\ (n-1)} \times \frac{1}{(a)^{n-1}}[2^{n-1} - 1]$$

OR $\boxed{t_{\frac{1}{2}} = \frac{1}{k\ (n-1)} \times \frac{1}{(a)^{n-1}}\left[2^{n-1} - 1\right]}$ **OR** $\boxed{t_{\frac{1}{2}} \propto \frac{1}{(a)^{n-1}}}$

Therefore $t_{\frac{1}{2}}$ for nth order reaction depends on the initial concentration of the reactants.

2.For first order reaction; $t_{\frac{1}{2}} \propto \frac{1}{(a)^{n-1}}$

substituting n=1 ,we get $t_{\frac{1}{2}} \propto \frac{1}{(a)^{1-1}}$ i.e $\boxed{t_{\frac{1}{2}} \propto 1}$

therefore $t_{\frac{1}{2}}$ for a first order reaction is independent of initial concentration.

3.For second order reaction; $t_{\frac{1}{2}} \propto \frac{1}{(a)^{n-1}}$

Substituting n=2 we get , $t_{\frac{1}{2}} \propto \frac{1}{(a)^{2-1}}$

i.e $t_{\frac{1}{2}} \propto \frac{1}{a}$

therefore $t_{\frac{1}{2}}$ for the second order reaction is inversely proportional to to the initial concentration of the the reactants.

Consider a second order reaction in which [A]=[B] or a=b .the velocity constant for this reaction is

$$k = \frac{1}{t} \times \frac{c}{a(a-c)}$$

Substituting t= $t_{1/2}$ and c= $\frac{a}{2}$

We get , $k = \frac{1}{t1/2} \times \frac{\frac{a}{2}}{a(a-\frac{a}{2})} = \frac{1}{t1/2} \times \frac{1}{a}$

$$t_{\frac{1}{2}} = \frac{1}{k} \times \frac{1}{a}$$